

MCP Theory Questions

1. Explain the difference between a tool, a resource, and a prompt in MCP, with a one-line example of when you'd reach for each

- **TOOL:** A tool is a function that allows an LLM to interact with external systems and access data that is not in its training data.

Example: A tool to send an email to someone.

- **RESOURCE:** A resource is a source of information/data that the AI can read and use as context.

Example: A resource that fetches today's weather information from a live API.

- **PROMPT:** A predefined instruction that helps the AI perform a particular task.

Example: "Summarize this research paper in 3 bullet points."

2. A local MCP server usually talks to Claude Desktop over stdio instead of a normal HTTP API. What does that mean in practice, and why does it make sense for this use case?

A local MCP server usually communicates with Claude Desktop using **stdio** (**standard input/output**) instead of HTTP.

- **stdio (Standard Input and Output):** It means Claude Desktop starts the MCP server locally and communicates with it through standard input and output. Claude sends requests to the server through **stdin**, and the server sends responses back through **stdout**.
- This makes sense because the MCP server runs on the same computer as Claude Desktop. Stdio is simple, fast, and avoids unnecessary network setup. It also helps Claude access local files, programs, and other resources.